

BUBBLE SORT PROGRAM

```
#include <stdio.h>

int main()
{
    int i, n, temp, j, arr[10];
    printf("\n Enter the number of elements in the array : ");
    scanf("%d", &n);
    printf("\n Enter the elements: ");
    for(i=0;i<n;i++)
    {
        printf("\n enter the arr[%d] element : ",i+1);
        scanf("%d", &arr [i]);
    }
    for(i=0;i<n;i++)
    {
        for(j=0;j<n-i-1;j++)
        {
            if(arr[j] > arr[j+1])
            {
                temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
    printf("\n The array sorted in ascending order is :\n");
```

```
for(i=0;i<n;i++){  
    printf("%d\t", arr[i]);  
}  
}
```

OUTPUT:

Enter the number of elements in the array : 5

Enter the elements:

enter the arr[1] element : 4

enter the arr[2] element : 3

enter the arr[3] element : 2

enter the arr[4] element : 5

enter the arr[5] element : 1

The array sorted in ascending order is : 1 2 3 4 5

ALGORITHM:

Step 1: Repeat Step 2 For 1 = to N-1

Step 2: Repeat For J = to N - I

Step 3: IF $A[J] > A[J + 1]$

 SWAP $A[J]$ and $A[J+1]$

 [END OF INNER LOOP]

 [END OF OUTER LOOP]

Step 4: EXIT